

Aditya Tawde

AI Engineer

+91 9699880183 | adityatawde9699@gmail.com

www.linkedin.com/in/aditya-s-tawde | <https://adityastawde.vercel.app/>

PROFESSIONAL SUMMARIES

I am an Artificial Intelligence and Data Science student with a strong interest in machine learning, data analytics, and intelligent systems. I am passionate about leveraging data-driven approaches to solve real-world problems and continuously expanding my knowledge of emerging technologies in AI.

Through academic projects and self-directed learning, I have developed skills in programming, data analysis, and problem-solving. I enjoy exploring concepts such as machine learning, deep learning, data visualization, and software development while building practical solutions that bridge theory and application.

EDUCATION

06.2024 - Present **B.Tech in AI & Data Science**

MGM's Jawaharlal Nehru Engineering College (JNEC), Chh. Sambhajinagar, India

- Pursuing a Bachelor's degree in Artificial Intelligence and Data Science, with coursework focused on machine learning, data analytics, statistics, programming, and intelligent systems.

SKILLS

- **Languages** - Python, JavaScript, C/C++, HTML/CSS
- **Frameworks** - FastAPI, Django, Flask, React, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Transformers, OpenCV, Selenium, BeautifulSoup
- **AI / ML** - LLMs (LlamaCpp/Groq/Gemini), RAG (BM25+Dense+RRF), sentence-transformers, GGUF quantization, Whisper, NLTK, TF-IDF
- **Databases** - PostgreSQL, SQLite, Redis, Qdrant, SQLAlchemy, Alembic
- **Tools** - Git/GitHub, Docker, Streamlit, Plotly, Matplotlib, Vite, Firebase
- **Languages** - English, Hindi, Marathi

CERTIFICATIONS

Gen AI Foundational Models for NLP - IBM

Generative AI and LLMs: Architecture - IBM

Deep Learning & Neural Networks with Keras - IBM

Machine Learning with Python - IBM

Building GenAI-Powered Apps with Python - IBM

Developing AI Apps with Python & Flask - IBM

ML Foundations: Case Study Approach - UWash

Programming with JavaScript - Meta

PROJECTS

Cognate

[Link](#)

- Developed **Cognate**, a cross-platform, local-first AI productivity application using **Tauri**, **React**, **TypeScript**, and **SQLite** with a privacy-focused architecture.
- Engineered an **AI-powered task planning and scheduling system** that intelligently prioritizes tasks, manages deadlines, and optimizes daily workflows.
- Implemented an **offline-first architecture** with secure local data storage, delivering fast performance and ensuring user privacy without cloud dependency.

Amadeus AI

[Link](#)

- Autonomous AI operating layer on Clean Architecture — transport → app → core → infra — with plan-driven cognitive runtime and persistent execution graph (Plan → PlanStep → Observation → Reflection).
- LLM routing chain: LlamaCpp (local GGUF) → Groq (Llama 3.3 70B) → Gemini 2.5 Flash; Redis-backed daily quota tracking; LOCAL_ONLY_MODE for 100% on-device privacy on 4 GB RAM / CPU-only hardware.
- 70+ sandboxed tools across system, network, filesystem, productivity; hot-pluggable plugin system, HITL approval gate for destructive actions, JWT auth, prompt-injection resistance.
- Hybrid RAG engine: BM25 + dense vector (Qdrant) with metadata-enriched chunking; voice pipeline via Whisper STT → LLM → Edge TTS with real-time SSE streaming.

Ledger — AI-Native Personal Finance Platform

[Link](#)

- Multi-provider LLM fallback router: Groq → Gemini → Cerebras → Mistral; receipt OCR (PaddleOCR), AI auto-categorization, SSE-streamed advisor chat, PWA offline support.
- 35+ REST endpoints — transactions, budgets, portfolios (stocks/MF/crypto/FD/gold), GST computation (Indian tax), Tally XML export, credit health scoring (300–900), community benchmarks.
- Deployed on Vercel + Render free tier; solved Render cold-start via staggered GitHub Actions + Vercel cron ping layers.

Amadeus-chat — Local LLM CLI

[Link](#)

- 100% on-device inference using GGUF-quantized models (Qwen3.5-2B Q4_K_M); runs on 4 GB RAM, CPU-only with zero cloud dependency.
- Hybrid RAG: BM25 + dense retrieval fused via Reciprocal Rank Fusion (RRF), cross-encoder re-ranking, MD5 deduplication; ingests PDF, MD, CSV, JSON, HTML, PY.
- Sliding window + LLM-powered memory compression; token budget guard (warn at 80%, hard-block at 95%); whitelisted safe shell execution via psutil.

ACHIEVEMENTS

- Pitched Printhub at E-Summit 2026 Udyam Startup Pitching Competition, JNEC (Feb 2026)
- Concept2Code & Tech-Escape Room, Swayambhu 2026 Technical Fest, JNEC (Feb 2026)
- Shipped functional app at Dreamflow Buildathon 2025 (ID: 2026H2S02DFLOW-P00078)